Machine Learning

26-03-26

Class Imbalance issue

- this issue occurs when some of the target classes in classification model have very few datapoints in the available data.

- As model sees very few examples of such classes during training, its not able to learn them properly.

- In testing as well, we'll not be able to test the model fairly for such minority classes

eg. 100 datapts available

90 A + 10 B

train: 75 A + 5 B

test: 15 A + 5 B

test accuracy

Y_{act} = [A A A A A A A A A A A A A A A B B B B B]

Y_{pred} = [A A A A A A A A A A A A A A A A A A A A]

$$\text{accuracy} = \frac{15}{20} = 0.75$$

- We'll see how to handle 'class imbalance issue' for correct model training later.

- To get correct model performance in this case, we need to compute class level performance metrics such as precision / recall / f1 score.

- Simple accuracy is sometimes not alligned with business goal or outcome.
  In those cases as well, above metrics help to align model with business goals.

- precision / recall / f1 → allign nicely with business outcomes.

- Using confusion matrix, we can easily compute prec / rec / f1

Confusion Matrix

height
weight → $\boxed{\text{binary clf}}$ → T / E

In binary clf, one of the classes is called +ve class and other one as -ve class

lets say here

T → +ve → 1
E → -ve → 0

} we can also set E → +ve and T → -ve as it doesn't change the result.

Y_{act} = [T T E T E E T E E T]

Y_{pred} = [T T T E E T T E T T]

Sr.no.	1	2	3	4	5	6	7	8	9	10
Y_{act}	T	T	E	T	E	E	T	E	E	T
Y_{pred}	T	T	T	E	E	T	T	E	T	T

$$\text{accuracy} = \frac{6}{10} = 0.6$$

T → +ve class E → -ve class

True positives (TP) = no. of examples where Y_{pred} is +ve and Y_{act} is +ve
= 4 (Sr.no. → 1, 2, 7, 10)

False positives (FP) = no. of examples where Y_{pred} is +ve and Y_{act} is -ve
= 3 (sr no. 3, 6, 9)

True Negatives (TN) = no. of examples where Y_{pred} is -ve and Y_{act} is -ve
= 2 (sr.no. 5, 8)

False Negatives (FN) = no. of examples where Y_{pred} is -ve and Y_{act} is +ve
= 1 (sr. no. 4)

True positives

True means $Y_{pred} = Y_{act}$

positives means Y_{pred} is +ve class

False positives

False means $Y_{pred} \neq Y_{act}$

positives Y_{pred} is +ve class

True Negatives

True means $Y_{pred} = Y_{act}$

Negatives means Y_{pred} is -ve class

False negatives

False means $Y_{pred} \neq Y_{act}$

Negatives means Y_{pred} is -ve class

confusion matrix

Y_{act} \ Y_{pred}	+ve	-ve
+ve	TP = 4	FN = 1
-ve	FP = 3	TN = 2

$$\text{confusion matrix} = \begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix}$$

$$\text{precision (+ve class)} = \frac{\text{no. of correct +ve predictions}}{\text{total no. of positive predictions}}$$

$$\frac{TP}{TP+FP} = \frac{n[Y_{pred} \text{ is +ve and } Y_{act} \text{ is +ve}]}{n[Y_{pred} \text{ is +ve}]}$$

$$\text{precision (+ve class)} = \frac{TP}{TP + FP}$$ } more popular formula

$$\text{precision (-ve class)} = \frac{TN}{TN + FN}$$

precision tells how precise model is when it predicts any class

Recall

$$\text{recall (+ve)} = \frac{\text{no. of correct +ve predictions}}{\text{total no. of +ve examples in } Y_{act}}$$

$$= \frac{n[Y_{pred} \text{ is +ve and } Y_{act} \text{ is -ve}]}{n[Y_{act} \text{ is +ve}]}$$

$$\text{recall (+ve)} = \frac{TP}{TP + FN}$$

$$\text{recall (-ve)} = \frac{TN}{TN + FP}$$

recall tells how many examples of given class from Y_{act} model is able to correctly recall.

In real business related cases, sometimes precision is preferred over recall and vice versa.

Precision vs Recall

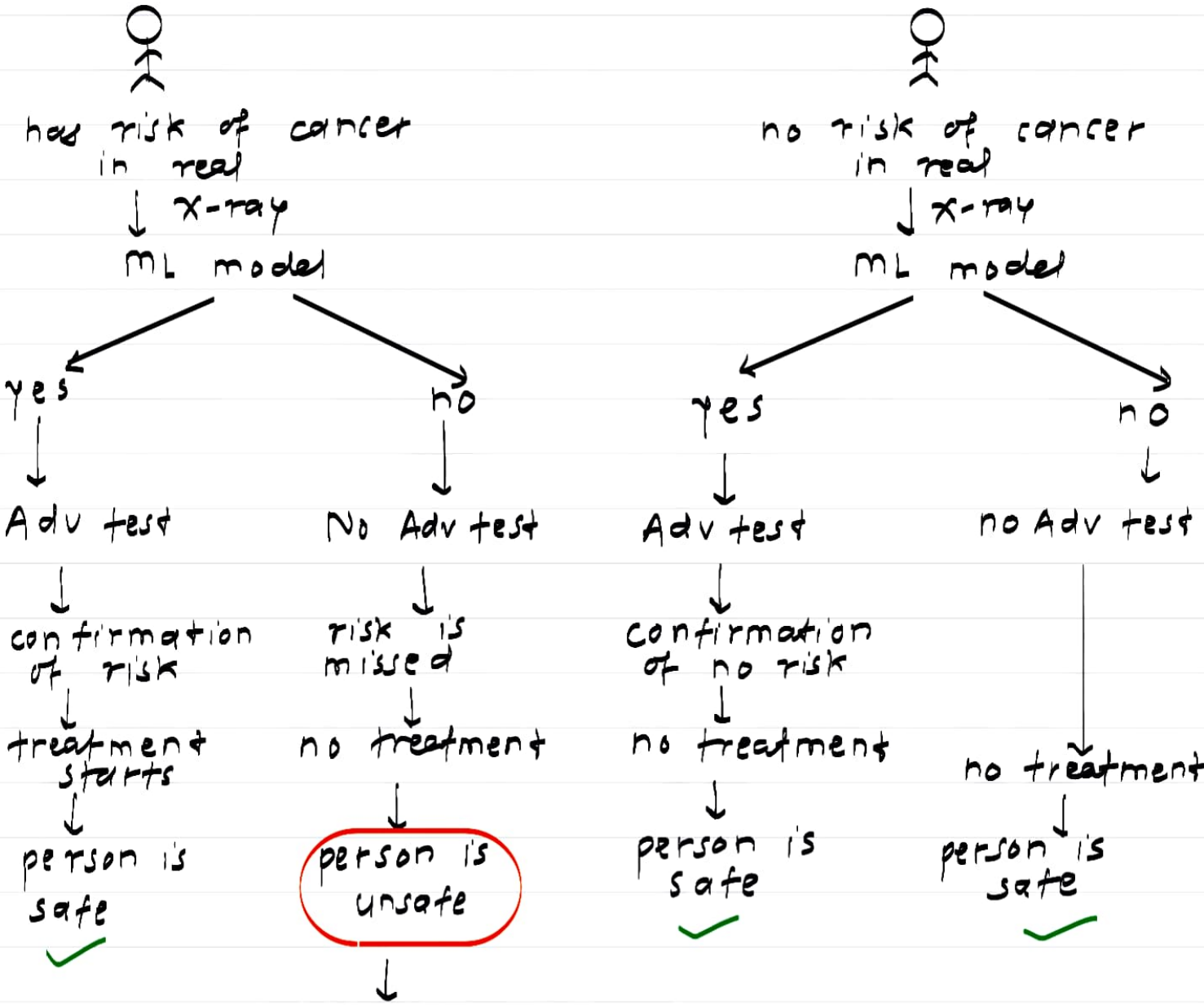
eg.

cancer detection

x-ray → $\boxed{\text{model}}$ → risk of cancer : yes/no

↓

binary clf



Yact is +ve → has risk

Ypred is −ve → no risk

in this model, we want to increase no. of correct predictions where Yact is +ve

in this case,

n (Y_{act} is +ve) = 10 → these many have risk in real

we want maximum of these 10 to be predicted as +ve by our model

Goal is →

if $\underbrace{Y_{act} \text{ is +ve}}_{TP + FN}$ increase cases where $\underbrace{Y_{act} \text{ is +ve and } Y_{pred} \text{ is +ve}}_{TP}$

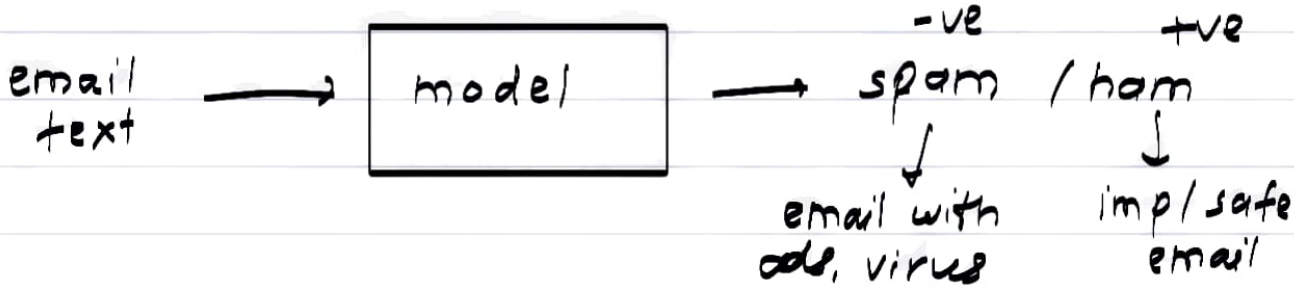
to meet above goal we need to focus more on recall of the model

$$\text{recall} = \frac{TP}{TP + FN}$$

in this we case

model 1	model 2
prec = 0.85	prec = 0.98
rec = 0.92	rec = 0.91
✓	✗
if 100 people have risk of cancer, model is able to find 92 of those	if 100 people have risk of cancer, model is able to find 91 of those
8 people missed and are unsafe	9 people are missed and are unsafe

this is the case where <u>recall</u> is preferred over all other metrics.

eg.

Email spam classification